

Decentralized Ledger Technology for EPCIS 2.0: Utilizing NFTs for Enhanced Product Traceability

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Abstract—After the release of version 2.0 of the Electronic Product Code Information Services (EPCIS), the standard consolidated as one of the most prominent standards for supply chain traceability in many industries. In this work, we introduce an innovative framework that integrates the EPCIS 2.0 standards with Hyperledger Fabric Decentralized Ledger Technology (DLT) to significantly improve product security and traceability within supply chains. By designating each product as a Non-Fungible Token (NFT), we establish a direct link between EPCIS events and NFT operations, enabling precise traceability and enhanced security. Our method maps standard events to token operations and the implementation of a traceability module employing hash mapping to manage product lifecycle information efficiently. In addition, we present algorithms designed to prevent counterfeiting by verifying product originality through EPCIS event data analysis. We evaluate the model considering relevant criteria, like performance on capturing and querying traceability information using Hyperledger Caliper, and show that our traceability query improves query time significantly without much overhead in capturing.

Index Terms—EPCIS 2.0, Traceability, Blockchain, DLT, Decentralized Network

I. INTRODUCTION

The increasing use of digital technologies is transforming the way supply chains operate. Technologies such as the Internet of Things (IoT), artificial intelligence (AI), and blockchain are helping companies to optimize their operations, reduce costs, and improve efficiency [1], while technologies such as RFID tags, GPS tracking, and real-time data analytics are helping companies to gain greater visibility into their supply chains [2]. Companies perceive an evolving demand for sustainable practices, quality of information, and transparency in the supply chain segment, and the need for prompt response to unexpected events, such as the recent COVID-19 pandemic [3]. Counterfeiting is also an important issue in supply chains, particularly in industries such as pharmaceuticals, luxury goods, and electronics, which causes billions of dollars in lost revenue for companies and is potentially harmful to consumers due to poor quality and lack of safety standards compared to genuine products.

Recently, the European Union started an initiative to work on a Digital Product Passport (DPP), with the aim of increasing the traceability of products and promoting sustainable practices, such as recycling, improved repair, and remanufacturing [4]. In the USA, the Drug Supply Chain Security Act (DSCSA) established, from November 2023, the mandatory use of electronic traceability systems for prescription drugs as they are distributed from manufacturers to pharmacies [5]. Such regulations increase the necessity for traceability solutions, with reliable information. Standards such as Electronic Product Code Information Services (EPCIS) and Core Business Vocabulary (CBV), promoted by the GS1 Organization, provide a common language and framework for data exchange, allowing companies to exchange information with supply chain partners in a standardized way [6].

Decentralized network technologies, such as Blockchain and Decentralized Ledger Technologies (DLT), represent a promising solution for supply chain traceability, as such technologies implement cryptographic validation of data and transactions, which have strong synergy with the DPP concept. Token standards such as ERC-20, ERC-721, and ERC-998 can represent real-world products as digital assets [7], thus establishing the basis for the Digital Product Passport (DPP) [4], and enabling the operations of such products in the digital economy with scalable, auditable, and secure methods, in accordance with market standards for electronic product information.

This work proposes an architectural model and implementation on how to integrate EPCIS 2.0 standards with blockchain technology. The model represents products as digital assets in the form of Non-Fungible Tokens (NFT), having all EPCIS event types and actions mapped to corresponding token operations (mint, transfer, burn), and aggregation events by the adoption of the ERC-998 standard. We implemented the model using Hyperledger Fabric and evaluated performance results (transactions per second and latency) using Hyperledger Caliper [8].

II. BACKGROUND AND RELATED WORK

A. EPCIS 2.0

Standards such as Electronic Product Code Information Services (EPCIS) and Core Business Vocabulary (CBV), promoted by the GS1 Organization, provide a common language and framework for sharing data, allowing companies to exchange information with supply chain partners in a standardized way. In June 2022, GS1 released version 2.0 of EPCIS/CBV, bringing significant updates to the previous version from 2016. The new version provides support to sensor data, certification details, linked organizations and locations, browsable online definitions, JSON/JSON-LD syntax, REST API for capture and query of event data, and GS1 Digital Link URI syntax to express GS1 identifiers within EPCIS event data [6].

The EPCIS/CBV 2.0 standard specifies five types of events. Object events represent every time an object is observed. Aggregation events document activities such as packing/unpacking and loading/unloading. Transformation events represent whenever products undergo a permanent change. Transaction events are recorded for the (dis)association of objects with business transactions, capturing physical relationships between objects when contained within one another. Each event comprises five critical dimensions—What, When, Where, Why, and How—which, respectively, identify the entities involved, record the timing, indicate the location, describe the business process activity, and provide sensor information about the products.

B. Decentralized Networks and Non-Fungible Tokens

Decentralized network technologies utilize ledger structures, distributed systems, consensus algorithms, and cryptographic mechanisms to improve transparency, data immutability, consistency, equal rights, and data availability [9]. Transactions, which are the cornerstone of data processing and sharing in these networks, are aggregated into blocks that must adhere to specific consensus rules for acceptance by peers [10]. The security of these transactions hinges on advanced cryptographic techniques such as Asymmetric Key Pairs, Hashing Algorithms, and structures such as Merkle and PATRICIA trees, supplemented by Zero-Knowledge Proofs and Homomorphic Encryption [11]. Such networks contain peered nodes exchanging data, with the transaction finality dependent on the chosen consensus algorithm. These networks are typically divided into public blockchains, where any node can participate, and permissioned Distributed Ledger Technologies (DLT), which are governed by rules dictating node participation and often structured around consortia [12].

Central to the functionality of these networks are tokens, which represent values transacted between cryptographic key holders. Tokens can be fungible, where each unit is identical to another, or non-fungible (NFT) [13], where each token is unique, adding to the complexity and utility of the network. Semi-fungible tokens, which represent a hybrid model, have a limited amount of identical units. The industry has set

various operational standards for these tokens, mainly through Ethereum Requests for Comments (ERC). The most popular standards include ERC-20 for fungible tokens, ERC-721 for non-fungible tokens, and ERC-1155 for semi-fungible tokens. The introduction of ERC-998 is particularly noteworthy as it allows an ERC-721 token to own another ERC-721 token, facilitating operations such as token bundling. These standards are implemented via Smart Contracts, programmable codes that run on each network node to process transactions according to specific business rules, thereby expanding the possible applications, including supply chains.

C. Related Work

Most related work on the keywords “epcis, blockchain, traceability” still adopts EPCIS version 1.2 or does not adopt any standard for product information data. As the volume of data in traceability systems tends to reach thousands of registries for each product, some related works adopt mixed storage strategies, including on-chain and off-chain data combined. Ahead, we present related works that are relevant to combining such elements.

The work of Li et al. [14] presents an approach for EPCIS standard adoption combined with DLT to approach product traceability. The model proposes the use of a combination of on-chain and off-chain data strategy, where EPCIS data relies on a MongoDB instance (off-chain), while a hash of each corresponding EPCIS registry is stored on-chain by a chaincode (Hyperledger Fabric’s implementation of Smart Contracts). Thus, as there is no EPCIS event semantic inside the chaincode rules, all the logic relies on an off-chain mechanism, whose data availability and protection mechanisms are not in the manuscript. The chaincode additionally assumes the responsibility of furnishing device identification upon organizations’ requests and validating these IDs when prompted. The authors propose that this validation occurs sporadically to alleviate the system’s burden. Given that the model was trialed in a setting comprising three organizations and one orderer, the potential repercussions of the ID verification process in a production-grade network remain uncertain.

The work of [15] applies EPCIS standard to the Food Supply Chain. The authors propose a combination of on-chain and off-chain data storage, to reduce storage occupation and the distribution of sensitive data on-chain. A set of Ethereum Smart Contracts manage permission verification and provide access to centralized off-chain services after checking requester’s permission. The work does not provide information on off-chain data protection, such as data encryption. They tested the model with a data set of up to 9GB of on-chain data. It does not provide information on the number of products and events, nor the payload size for each on-chain data. Yele and Litoriya [16] also propose a method for food traceability, specifically for restaurants, adopting Hyperledger Fabric DLT. The method does not adopt any specific data standard for product information and assumes that data from a batch of products will be available in the DLT, allowing further steps in the supply chain to read information about such a batch

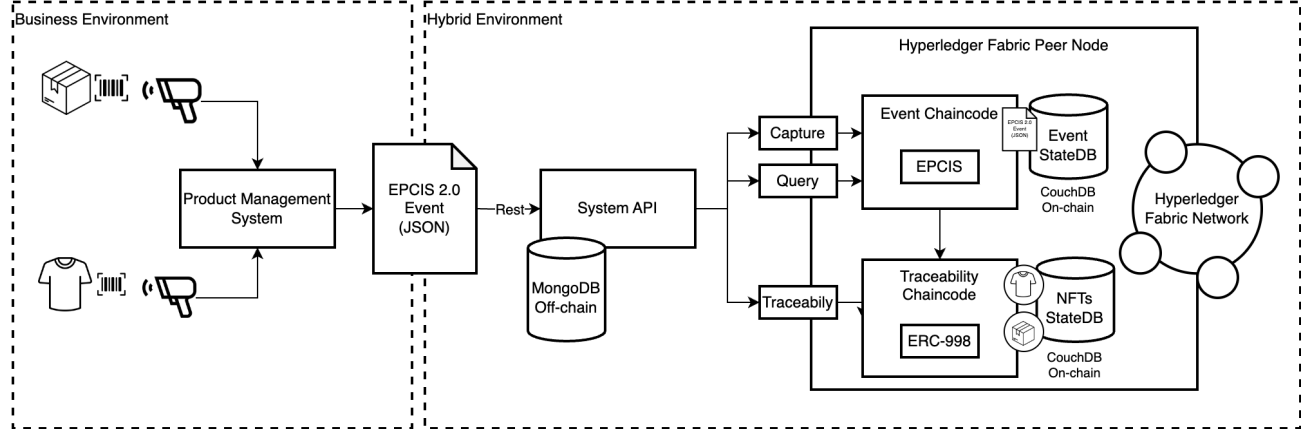


Fig. 1: Proposed Model architecture overview: the Business Layer comprises client applications, while the Organization's Premises will keep the core solution elements, that replicate data to other authorized organizations via Hyperledger Fabric channels.

via a QR Code. They evaluate the model in a set of fifteen personal computers, focusing on the impact on performance (blocks per second) as the number of nodes and orderers in the Hyperledger Fabric network changes. Results reached 10 blocks per second in a scenario with only one endorsing node and one orderer node. There is no description of the data set used for experiments.

In [17], EPCIS standard, distributed networks, and Internet of Things (IoT) sensors combine for supply chain sensing security. The work analyses different tracking and monitoring aspects of sensing and proposes a set of goals to address issues related to data trustworthiness and information security: tamperproofness, authenticity, completeness, affordability, and latency agnosticism. It considers the supply chain a low-trust scenario and proposes methods to tamper-proof sensor data among participating actors in a given supply chain. Four scenarios are part of the analysis, with different security levels each. They evaluated the model using Quorum private blockchain in a network of four servers. The network stores only data hash and reaches a performance result of 50 transactions per second.

Dietrich et al. [7] propose a model to manage objects as tokens in a blockchain. The model explores existing standards and introduces an evolution of the ERC-1155 standard called Blueprint that allows for units in the same token to have different IDs and proposes a set of Smart Contracts and Decentralized Apps (DApps) to manage the governance of roles and permissions. Each token operation is mapped to a corresponding EPCIS event, as follows: Object Event Create corresponds to Mint, Transaction Event corresponds to Transfer, Aggregation, and Disaggregation corresponds to operations in a proposed element called Token Bucket, which is a mechanism that manages token memory to keep track of which tokens aggregate another. This approach differs from our proposal, as we adopt the already existing ERC-998 standard to manage such events and the newer version of

EPCIS.

III. PROPOSED MODEL AND IMPLEMENTATION

To enhance product security and traceability within the supply chain, we have proposed and implemented an architectural framework that merges EPCIS 2.0 standards with blockchain technology. Our approach uniquely designates each product as a Non-Fungible Token (NFT), thereby facilitating the direct association of EPCIS events and actions with specific NFT operations.

This methodology involves recording traceability data by attaching a hash reference of event data to each product tokenized as an NFT. The comprehensive architecture is illustrated in Figure 1. As depicted, each occurrence of a new event prompts the event chain code to store the corresponding EPCIS event in the state database and initiates the traceability chain code to process the event. This traceability component then engages in NFT operations to generate NFTs and compile traceability information.

A. Network Architecture

The proposed model adopts the model of a permissioned network as underlying technology. The group of participants in the network might include manufacturers, retailers, logistic operators, and government bodies, depending on the industry and use case. The main incentive for network participants is the construction of a shared tamper-proof base of data and transactions for products and events in a standardized format.

We adopted Hyperledger Fabric as the solution to run the permissioned network. It implements two types of nodes: peer nodes to sign transactions and keep state databases for Smart Contracts, and ordering nodes to reach consensus on transactions and keep ledger databases for each channel, which is a subgroup of organizations in the network. Each organization can provide their own certification authority mechanism to provide credentials to their users.

B. EPCIS Event Representation through NFT Operations

To ensure comprehensive coverage of all EPCIS standard events, including ObjectEvent, AggregationEvent, TransactionEvent, TransformationEvent, and AssociationEvent, we have carefully mapped these standard events and their respective actions to specific token operations. This mapping enables us to represent every product (EPCs) as a non-fungible token (NFT), linking event types and actions to corresponding token operations in a nuanced manner. The link with physical product is the product unique identification. Any sensitive information regarding products might rely outside the distributed network while still referencing the same product identification.

Specifically, for ObjectEvent types paired with an "Add" action, we employ a Token Mint operation to signify the creation or addition of a new product. Conversely, when an ObjectEvent type is associated with a "Delete" action, we utilize a Token Burn operation, indicating the removal or deletion of a product from the system. For TransactionEvent types that involve an "Add" action, a Token Transfer operation is executed, reflecting the change in ownership or the movement of products. Similarly, AggregationEvent types with an "Add" action are handled through Token Transfer operations, but uniquely, these operations facilitate the transfer of tokens from aggregated EPCs (individual products) to an aggregating EPC (a collective grouping), effectively illustrating the aggregation process.

One of the major contributions of our model is the adoption of the ERC-998 Composable NFT token standard to represent EPCs. Particularly, to address the nuances of Aggregation Events, we take advantage of its ability to allow one token ID to own another token ID. This capability is ideally suited to aggregation operations within the EPCIS framework, as it accurately depicts the hierarchical relationship between individual products and their collective groupings. Furthermore, this standard imposes restrictions on the transfer of a given EPC while it is aggregated to another EPC, thereby enhancing the system's ability to maintain product integrity and traceability throughout the supply chain. Figure 2 shows the interaction of smart contracts to create NFTs.

C. Optimizing Product Traceability in EPCIS: A Hash Mapping Solution

Tracing a product through the EPCIS framework can become complex, requiring multiple queries and processing steps, especially when products undergo aggregation and transformation events. Once a product is aggregated, subsequent events record only the parent ID, complicating direct traceability through queries with the product's ID alone. This necessitates a series of recursive queries for accurate traceability, a process that becomes similarly intricate during transformation events, where a product is converted into a different item with a new unique product ID.

To address these challenges, we developed and implemented a traceability module designed to maintain comprehensive traceability information throughout EPC token operations. As illustrated in Figure 3, our analysis led to the identification

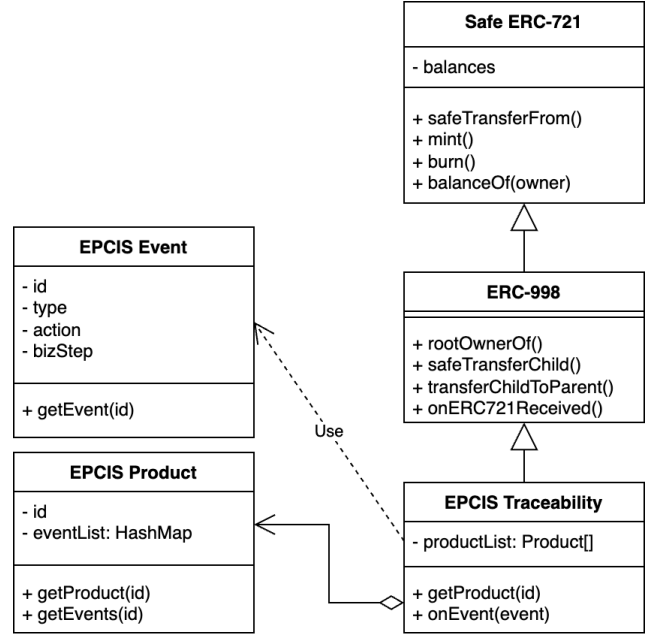


Fig. 2: Smart contracts for Capture and Query Interfaces, and for Product Traceability, with the adoption of token standards to represent items.

of three distinct strategies for enabling product traceability: Traceability Calculation, Data Replication, and Hash Mapping.

- **Traceability Calculation** involves traversing the entire database to identify records related to the requested EPC, which may result in longer processing times and potentially affect the end user's experience, especially with large event datasets.
- **Data Replication** entails maintaining a separate storage system for traceability events associated with each EPC, effectively reducing processing time for retrieving traceability information but requiring additional storage space. This approach introduces the challenge of validating replicated traceability data, necessitating mechanisms to confirm the accuracy of replicated data against the original dataset.
- **Hash Mapping** offers a solution by storing traceability data as hash references to Object Events within the original dataset. This method demands less storage space than data replication and circumvents the issue of data duplication. The traceability data serves as a structured reference to Object Events, utilizing a unique ID (the hash) as the key for mapping. Techniques such as the Merkle Patricia Trie, used in the Ethereum network, can be applied for efficient data management and verification.

Our work employs the technique of hash mapping to construct a reliable and efficient traceability framework. For every new EPCIS event, the algorithm will add the event hash to a map in every product it relates, controlled by the Smart Contract.

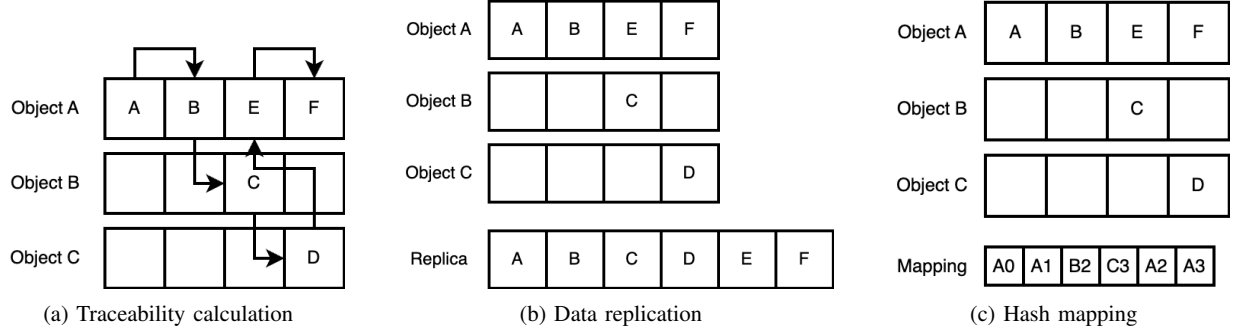


Fig. 3: Product traceability strategies: traverse event history on-demand to produce traceability data reduces storage occupation but incurs high response time, while replicating event data reduce processing time drastically with a high demand for storage. The Hash Mapping solution is more efficient in terms of storage occupation and reduces the processing time for history traversing.

This approach allows for streamlined traceability of products throughout their lifecycle in the supply chain, from production through various transformations and aggregations. By utilizing hash mapping, we facilitate quick and accurate access to a product's history and transformations without demanding extensive database searches or complex query mechanisms. This method ensures the integrity and verifiability of the traceability data and significantly reduces the storage requirements and processing times associated with traditional traceability methods. Through this innovative approach, we provide a scalable and robust solution for enhancing product traceability within the EPCIS framework, leveraging the power of blockchain technology to secure and streamline supply chain operations.

D. Counterfeit Module

To help prevent counterfeiting, we propose multiple algorithms responsible for product originality analysis based on an EPCIS event data set. The input data will be analyzed against records in the system to find matching products. In case of any mismatching input data, the module will return an error message, alerting of a potential fraud. For example, an `ObjectEvent` with an `ADD` action must occur for a given product before any other event type. Therefore, any event captured about a product before such an event will be considered fraud.

Every incoming event undergoes a set of rules that generates a score for each product in the event of being counterfeit. Different industries can specialize the rule set according to their needs by extending the base Smart Contract. In Algorithm 1, we provide an example of such a rule, representing the aforementioned example.

IV. EVALUATION AND RESULTS

To assess the performance and reliability of our system, we implemented an evaluation using Hyperledger Caliper [8], an open-source benchmarking tool specifically designed for testing blockchain technologies such as Hyperledger Fabric. As depicted in Figure 4, Caliper employs several worker nodes to generate a significant workload, aiming to rigorously test

Algorithm 1 Counterfeit algorithm

```

procedure COUNTERFEITRULE(e: Event)
  if e.action == ADD then
    return 0
  end if
  productList ← getProducts(e.epcList)
  for all p ∈ productList do
    productEvents ← getEvents(p)
    for all pe ∈ productEvents do
      if pe.action == ADD then
        return 0
      end if
    end for
  end for
  return 1
end procedure

```

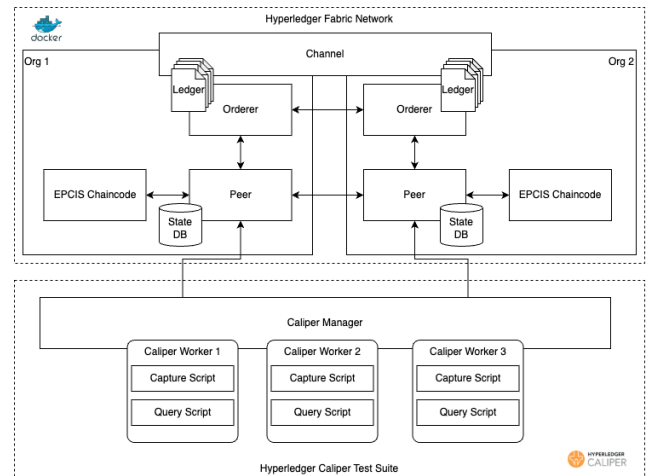
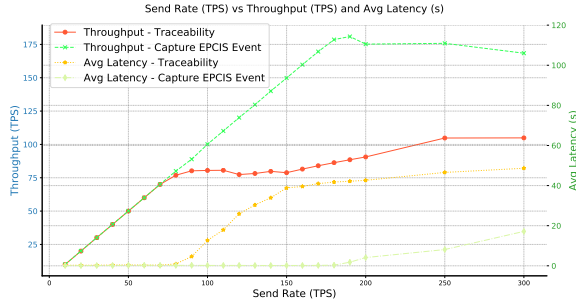
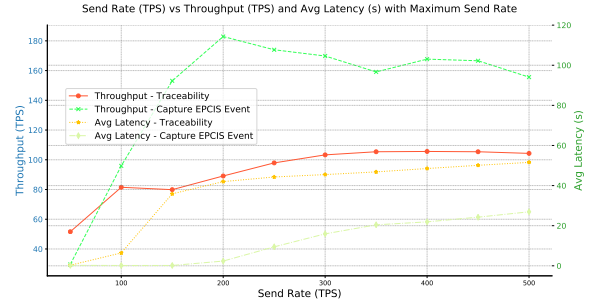


Fig. 4: Test configuration using a caliper to evaluate the system performance

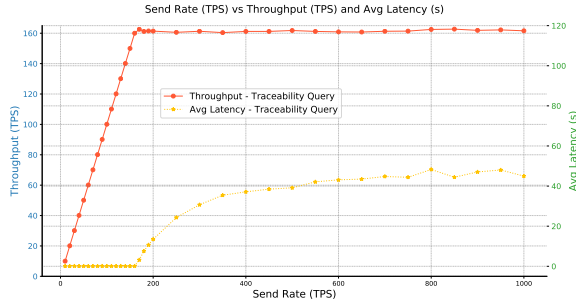


(a) A fixed rate is employed in the caliper configuration

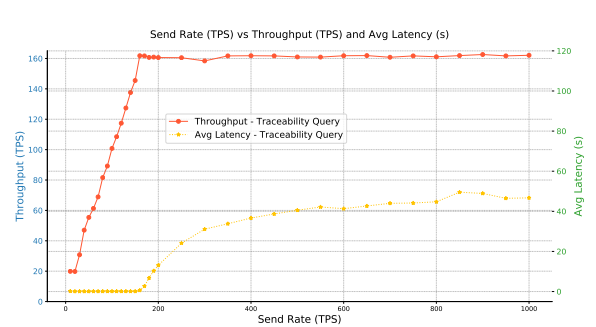


(b) A maximum rate is employed in the caliper configuration

Fig. 5: Comparison of capturing an event and capturing traceability, while varying the send rate.



(a) A fixed rate is employed in the caliper configuration with 30 number of workers.

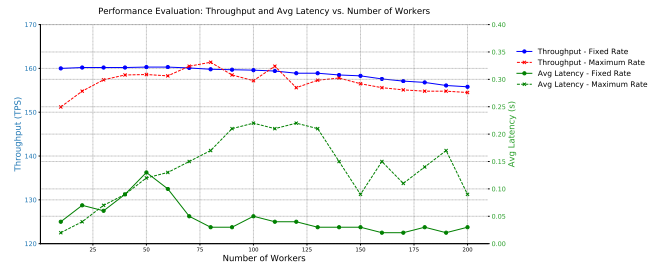


(b) A maximum rate is employed in the caliper configuration

Fig. 6: Comparison of tracing a product while varying the send rate. Here the product is aggregated only one time.



(a) Performance evaluation of traceability event capturing with a variable number of workers. In this setup, a constant TPS rate of 100 is maintained in the caliper while varying the number of workers configuration to evaluate system performance, specifically focusing on throughput and average latency, under controlled high-performance conditions



(b) Performance evaluation of traceability query with a variable number of workers. In this setup, a constant TPS rate of 160 is maintained in the caliper configuration while varying the number of workers to evaluate system performance, specifically focusing on throughput and average latency, under controlled high-performance conditions.

Fig. 7: Comparison of capturing and querying traceability while varying the number of workers at a fixed transaction per second (TPS) rate.

the capabilities and resilience of the system under evaluation. This process involves the creation of a simulated environment where Caliper actively monitors the system's response to various stress conditions. To run both the test network and Caliper, our setup comprised a robust hardware configuration, including an Intel processor with 12 cores and 46GB of RAM, running on an Ubuntu 20.04 system.

To evaluate the capturing performance, we conducted tests for capturing a simple EPCIS event and then an event with traceability processing. We tested the system under varying send rates (transactions per second, TPS) in two scenarios: a fixed-rate scenario, where Caliper workers maintain a constant push rate to intentionally stress the system, and a maximum-rate scenario, where they push at the highest sustainable rate without causing system overload. The results, depicted in Figures 5a and 5b, show that capturing traceability events results in lower performance than capturing simple events. This reduction in efficiency is due to the additional operations required for NFT tokenization and hash mapping.

Differently from [7], our method adopts standard ERC-998 to support aggregation events, instead of a custom method, which elevates long-term support to the method. The evaluation results reached 150 average TPS that, compared to [17], is three times higher. As most related work supports only version 1.2, supporting EPCIS 2.0 is a significant contribution, due to its additional functionalities, and support to widely adopted industry standards.

Regarding query performance, we have carried out experiments to assess how the system handles traceability queries. We present the result in figure 6. Under both maximum-rate and fixed-rate configurations, the system demonstrated a consistent throughput capacity, stabilizing around 160 to 163 transactions per second (TPS), regardless of increases in send rates up to 1000 TPS. However, both configurations exhibited a significant rise in average latency once exceeding the 160 TPS threshold, with latency increasing sharply to approximately 45 seconds at higher send rates. It indicates that while the system can sustain a high throughput up to a specific limit, its performance in terms of response time deteriorates significantly under higher transaction volumes, highlighting a critical trade-off between maintaining throughput and managing latency effectively. Compared to recursive queries, which have an average latency of 3.79 and throughput of 19.5 TPS under a send rate of 20 TPS, the traceability system significantly outperforms the recursive queries by performing around 160 TPS under a second. This performance is only when the level of aggregation is 2. The performance will be significant when there are more layers of product aggregation.

We also investigated how changing the number of workers from 10 to 200 affects traceability capture performance under both fixed-rate and maximum-rate sending scenarios at 100 transactions per second (TPS) for capture and 160 TPS for querying traceability information, as illustrated in Figure 7. In transaction capture, both under fixed and maximum-rate conditions, we observed that throughput decreases and latency increases as more workers are added. It suggests diminishing

returns and escalating costs as the system scales up. In contrast, during querying, the system demonstrates consistent performance in terms of throughput and latency, regardless of the number of workers.

V. CONCLUSION AND FUTURE WORK

This study presented a model integrating EPCIS 2.0 standards with the Hyperledger Fabric private blockchain, enhancing supply chain traceability and security by designating each product as a Non-Fungible Token (NFT). The framework effectively maps standard EPCIS events to specific NFT operations, ensuring that each product's journey within the supply chain is accurately recorded and securely maintained. Through our proposed traceability module and hash mapping techniques, we efficiently manage traceability information, significantly improving the query performance.

Performance evaluations using Hyperledger Caliper highlight the system's enhanced efficiency in traceability queries, demonstrating substantial improvements in query times with minimal impact on transaction overhead while capturing. Compared to related work, the proposed method has a meaningful contribution to adopting DLT technologies for traceability. The support to EPCIS 2.0 with NFT establishes the basis for implementing the Digital Product Passport concept.

For future work, we recommend a performance evaluation with a production-grade network, having multiple organizations and nodes communicating over the Internet. The counterfeit module needs more rules to address industry-specific demands. We also recommend that the system implements means to support direct ERC token operations, generating counterpart EPCIS events automatically. It allows the system to be compatible with traditional token solutions, such as cryptographic wallets and improves the user experience while keeping support to the EPCIS 2.0 standard.

VI. ACKNOWLEDGEMENT

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